

Satellite Data Visualization Platform for Weather and Climate Pattern Analysis

A Case Study Analysis Report

Submitted in the partial fulfilment for the Course of

CSA1012 - SOFTWARE ENGINEERING

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

AI & ML

Submitted by

Y.Navya(192425177)

Under the Supervision of

Dr.K.Anita Davamani



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

July- 2026

Table Of Contents

S.No	Title	Page No
1.	Understand the Problem Statement	3-4
2.	Identify Key Stakeholders and Challenges	5-7
3.	Analyze the Current Approach	8
4.	Propose a Suitable Solution Using Software Engineering Principles	9-12
5.	Justify the Chosen Methodology/Framework	13-14
6.	Discuss Expected Outcomes and Limitations	15-16

1. Understand the Problem Statement

1.1 Problem Understanding

Satellite imagery is widely used for weather forecasting, climate monitoring, disaster management, and environmental research. Organizations such as NASA and NOAA generate huge volumes of satellite data every day. However, this data is stored in scientific formats that require specialized software and technical expertise to process and interpret.

Most government agencies, researchers, students, and planners cannot easily access or understand this information. They often spend significant time converting raw satellite data into meaningful visualizations. Existing tools are expensive, technically complex, or lack integrated visualization features.

The proposed system aims to develop a web-based Satellite Data Visualization Platform that automatically collects satellite data, processes it, and presents it through interactive maps, graphs, and dashboards. This enables users with little technical knowledge to analyze weather conditions and long-term climate patterns.

1.2 Objectives of the Proposed System

The proposed system aims to:

1. Retrieve satellite imagery from NASA, NOAA, and other public sources.
2. Display real-time and historical weather information on interactive maps.
3. Analyze long-term climate trends using time-series charts.
4. Detect anomalies such as cyclones, droughts, floods, and heat waves.
5. Compare multiple environmental datasets (temperature, rainfall, vegetation, cloud cover).

6. Allow users to export processed datasets and reports.
7. Support region-wise filtering for localized analysis.
8. Provide an easy-to-use web interface for both experts and non-experts.

1.3 Scope of the Problem

The system covers:

- Satellite data collection
- Data preprocessing
- Interactive visualization
- Climate trend analysis
- Weather anomaly detection
- Regional filtering
- Data download and reporting
- Web-based access

The platform is intended for:

- Meteorologists
- Researchers
- Government agencies
- Disaster management authorities
- Agricultural planners
- Environmental organizations
- Students
- Policymakers

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders

Stakeholder	Role
Meteorologists	Analyze weather patterns and forecasts
Climate Researchers	Study long-term climate changes
Government Agencies	Develop climate-related policies
Disaster Management Teams	Monitor disasters and plan emergency responses
Agricultural Departments	Plan irrigation and crop management
Environmental Organizations	Monitor Environmental Changes
Students & Educational Institutions	Learning & Research
System Administrator	Maintain servers and databases
Software Developers	Develop and improve the platform
Data Providers (NASA, NOAA)	Supply satellite datasets

2.2 Roles and Responsibilities

Meteorologists

- Monitor weather systems
- Predict storms and rainfall
- Analyze cloud movements

Climate Researchers

- Study historical climate trends
- Perform environmental analysis
- Publish research findings

Government Agencies

- Use climate data for policy making
- Plan climate adaptation strategies

Disaster Management Teams

- Monitor cyclones
- Plan evacuations
- Respond to disasters

Agricultural Departments

- Monitor drought
- Predict rainfall
- Improve crop planning

Students

- Learn weather analysis
- Perform academic projects

Software Developers

- Build software modules
- Integrate APIs
- Maintain system performance

System Administrators

- Manage servers
- Ensure system availability
- Secure databases

2.3 Challenges Faced by Stakeholders

Stakeholder	Challenges
Meteorologists	Large amount of raw data to analyze
Researchers	Complex scientific formats
Government	Delay in accessing processed information
Disaster Teams	Need real-time visualization
Farmers	Limited access to climate insights
Students	Difficult software and technical knowledge
Developers	Handling huge datasets
Administrators	Storage and security management

3. Analyze the Current Approach

3.1 Existing System

Currently, satellite images are obtained from agencies like NASA and NOAA. Users download raw datasets and process them using specialized scientific software such as GIS tools, MATLAB, Python libraries, or remote sensing applications.

The process is mostly manual and requires technical expertise.

3.2 Strengths of Existing System

- Highly accurate satellite data
- Reliable scientific sources
- Supports detailed research
- Large historical datasets available
- Trusted by researchers worldwide

3.3 Limitations

- Complex scientific file formats
- Expensive software
- Requires technical expertise
- Time-consuming preprocessing
- Poor visualization
- No integrated comparison tools

- Difficult for non-experts

3.4 Problems and Gaps

- Lack of user-friendly interface
- Manual processing consumes time
- No automatic anomaly detection
- Poor accessibility
- Limited collaboration
- Separate tools for visualization and analysis

4. Proposed Software Solution

4.1 Proposed Solution

Develop a cloud-based web application that automatically downloads satellite data from NASA and NOAA APIs, processes the data, and provides interactive dashboards for visualization and analysis.

Users can easily explore weather conditions, compare historical trends, detect anomalies, and download reports without specialized software.

4.2 Major Functional Modules

Module 1: Data Collection Module

- Retrieve satellite data
- Connect with NASA/NOAA APIs
- Automatic updates

Module 2: Data Processing Module

- Clean raw datasets
- Convert scientific formats
- Generate usable datasets

Module 3: Interactive Visualization Module

- Interactive maps
- Weather layers
- Heat maps
- Climate dashboards

Module 4: Climate Trend Analysis Module

- Temperature trends
- Rainfall analysis
- Cloud cover analysis
- Seasonal comparison

Module 5: Pattern Detection Module

- Cyclone detection
- Flood monitoring
- Heatwave alerts
- Drought identification

Module 6: Comparison Module

Compare

- Rainfall vs Temperature
- Vegetation vs Rainfall
- Cloud Cover vs Temperature

Module 7: Export Module

- PDF Reports
- CSV datasets
- Images
- Charts

Module 8: User Management Module

- Login
- Registration

- Role management
- Access control

4.3 Functional Requirements

- User Registration/Login
- Retrieve satellite data
- Interactive maps
- Weather visualization
- Climate trend graphs
- Pattern detection
- Region filtering
- Data comparison
- Report generation
- Data export

4.4 Non-Functional Requirements

- High Performance
- Scalability
- Reliability
- Security
- Availability
- Maintainability
- Usability
- Fast response time
- Cross-platform compatibility

4.5 Software Engineering Principles

Modularity

Separate modules for data collection, processing, visualization, analysis, and reporting make the system easier to develop and maintain.

Scalability

Cloud-based architecture allows additional satellite sources and users to be supported without major redesign.

Maintainability

Well-structured code and modular components simplify updates, bug fixes, and feature additions.

Reliability

Data validation, automated backups, and fault-tolerant services ensure continuous operation and accurate information.

Usability

A simple web interface with interactive dashboards enables both experts and non-experts to use the platform effectively.

Security

Authentication, authorization, encrypted communication (HTTPS), and secure data storage protect user accounts and downloaded datasets.

5. SDLC / Framework Justification

Selected SDLC Model

Agile (Scrum)

Why Agile?

The project requires continuous improvement because satellite data sources, visualization needs, and user requirements evolve over time.

Agile enables frequent updates, quick feedback, and iterative development.

Advantages of Agile

- Faster development
- Continuous testing
- Easy feature updates
- Customer feedback
- Flexible requirement changes
- Better collaboration
- Lower project risk

Scrum Framework

Product Owner

Defines project requirements.

Scrum Master

Manages Agile process.

Development Team

Develops software modules.

Sprint

2–4 week development cycles.

Sprint Review

Evaluate completed work.

Sprint Retrospective

Improve future development.

How Agile Supports the Project

- New satellite APIs can be added incrementally.
- Visualization features can evolve based on user feedback.
- Bugs are identified and fixed early through continuous testing.
- Enables continuous delivery of new features without disrupting existing functionality.

6. Expected Outcomes and Limitations

6.1 Expected Outcomes

- Easy access to satellite data
- Faster weather analysis
- Better climate monitoring
- Improved disaster preparedness
- Reduced manual processing
- Better agricultural planning
- Increased academic research
- Support for government decision-making
- Scalable platform for future expansion

6.2 How the Solution Addresses Challenges

Challenge	Proposed Solution
Complex scientific data	Automatic data processing
Difficult visualization	Interactive dashboards and maps
Manual analysis	Automated pattern detection
Lack of accessibility	Web-based interface
Multiple data Sources	Integrated visualization platform

Time-Consuming workflow	Automated data ingestion and analysis
-------------------------	---------------------------------------

6.3 Risks and Limitations

- Dependence on internet connectivity
- API downtime from data providers
- Large storage requirements
- High computational resources for real-time processing
- Delay in updating satellite data due to source availability
- Initial development cost
- Performance challenges with very large datasets

6.4 Future Enhancements

- AI-based weather prediction models
- Machine learning for improved anomaly detection
- Mobile application support
- Real-time disaster alert notifications
- 3D satellite visualization
- Integration with IoT weather sensors
- Multi-language support
- Voice-enabled dashboard
- Predictive climate analytics
- Integration with additional satellite sources (e.g., ESA, ISRO)

Conclusion

The proposed Satellite Data Visualization Platform for Weather and Climate Pattern Analysis provides a comprehensive, user-friendly solution for transforming complex satellite data into meaningful insights. By integrating automated data ingestion, interactive visualizations, climate trend analysis, and anomaly detection within a scalable web-based platform, it addresses the limitations of existing systems. Using the Agile (Scrum) methodology ensures flexibility, continuous improvement, and easier maintenance, making the platform suitable for meteorologists, researchers, government agencies, disaster management teams, students, and policymakers. Overall, the solution improves accessibility, supports informed

decision-making, and establishes a strong foundation for future advancements in climate analytics.